

<u>CL-2006</u> <u>Operating System Lab</u>	<u>Revision Lab</u>
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National University of Computer and Emerging Science
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Lab Activity

1. Open a terminal and execute the following tasks:
 - Create a directory named 'your roll no' in your home directory.
 - Navigate into the newly created directory.
 - Check if the GNU C++ Compiler (g++) is installed; if not, install it using the appropriate package manager command.
 - Inside the 'your roll no' directory, create a C++ file named 'RevisionC.cpp' using the touch command and write a program to print prime numbers till n.
 - Confirm the existence of the 'RevisionC.cpp' file and display the result after successfully compiling the source file.
2. In a Linux terminal, using a single echo command, print the following output:
"Welcome to Linux, Linux is free, secure and open source".
Additionally, execute the following tasks:
 - Create a folder named 'osRevlab' on the Desktop.
 - Inside the 'osRevlab' folder, include two text files named 'YourFirstName.txt' and 'YourLastName.txt'.
 - Generate a third file named 'mergedfile.txt' within the 'osRevlab' folder, and ensure that 'mergedfile.txt' contains the combined content of 'YourFirstName.txt' and 'YourLastName.txt'.
 - Copy 'mergedfile.txt' to a new file named 'copiedfile.txt' in the 'osRevlab' folder.
 - Delete all files (YourFirstName.txt, YourLastName.txt, mergedfile.txt, copiedfile.txt) inside the 'osRevlab' folder.
3. Perform the following tasks:
 - Identify the command to list files in the current directory that have names starting with an uppercase letter.
 - Create three new directories named 'Reports', 'Projects', and 'Archive' using a single command if they don't already exist.
 - Copy files in the current directory containing the character string 'report' into the 'Reports' subdirectory.
 - Copy files in the 'Reports' subdirectory with names ending in '.txt' or '.docx' into the 'Projects' subdirectory.
 - Copy files in the 'Projects' subdirectory containing the character string 'notes' or 'misc' into the 'Archive' subdirectory.
 - Copy files starting with 'important_' from the 'Archive' subdirectory into the 'Backup' subdirectory. Move files starting with 'obsolete_' from the 'Archive' subdirectory into the 'Backup' subdirectory.
 - Delete all files containing the sequence 'obsolete' from the 'Backup' subdirectory.
4. Write a script that accepts a sentence and counts the number of words in it.

5. Write a program that reads a text file named 'input.txt' and counts the number of words in it. Implement error handling for file opening and reading operations. Utilize `open()`, `read()`, and `close()` system calls.
6. Develop a program that creates four child processes. Each child process executes a different command-line utility (`ls`, `ps`, `whereis`, `mkdir`) using the `execlp()` function. The parent process waits for all child processes to complete and then prints a message indicating their completion.
7. Develop a program that writes a string provided as a command-line argument to a text file named 'data.txt'. Implement error handling for file opening and writing operations. Utilize `open()`, `write()`, and `close()` system calls.
8. Create the following functions in separate files (using `.h` and `.c` files) `EvenCount`, `OddCount`, `TotalCount`.
 - `EvenCount` counts the number of even numbers in the list.
 - `OddCount` counts the number of odd numbers in the list.
 - `TotalCount` counts the total numbers in the list.Create a list of numbers or take it from the user in the main function. Now compile all functions using `makefile`.